

Arbitrary-Order Hu–Zhang Mixed Finite Element Methods for Density-Based Topology Optimization

Liang He¹, Chunyu Chen^{2,*}, Huayi Wei²

¹ *School of Mathematical Sciences, Dalian University of Technology, Dalian 116024, China*

² *School of Mathematics and Computational Science, Xiangtan University; National Center of Applied Mathematics in Hunan, Hunan Key Laboratory for Computation and Simulation in Science and Engineering, Xiangtan 411105, China*

Abstract. To overcome the inter-element traction discontinuity in stresses recovered from displacement elements and the volumetric locking of low-order displacement elements in nearly incompressible regimes, this paper establishes an arbitrary-order Hu–Zhang mixed finite element framework for density-based topology optimization on simplicial meshes, using symmetric stress and displacement as independent unknowns. Combining jump stabilization for lower polynomial orders with local corner relaxation, we formulate models for compliance minimization and volume minimization subject to local stress constraints, derive consistent adjoint sensitivities incorporating the design dependence of the right-hand side introduced by traction lifting, and implement the framework within FEALPy. Two-dimensional numerical examples show that both native higher-order and low-order stabilized formulations achieve their theoretically expected convergence rates, and the unified optimization workflow is applicable across all investigated discrete polynomial degrees. In standard compliance optimization, the mixed and displacement methods yield comparable topologies and compliance values; in nearly incompressible regimes, the mixed method achieves designs with lower compliance than linear displacement elements, whereas increasing the displacement polynomial degree narrows this gap. Local stress constraints are evaluated directly from the primary stress variables without stress recovery, with the discrete stress naturally preserving inter-element normal traction continuity.

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1 Introduction

Topology optimization (TO) provides a systematic numerical design tool for high-performance lightweight structures by optimizing the spatial distribution of material within a prescribed design domain. Since the continuum material distribution approach was introduced by Bendsøe [4],

*Corresponding author. *Email addresses:* heliang@dlut.edu.cn (L. He), chunyuchen@xtu.edu.cn (C. Chen), weihuayi@xtu.edu.cn (H. Wei)

density-based topology optimization, particularly represented by the Solid Isotropic Material with Penalization (SIMP) method, has been widely adopted due to its conceptual clarity, algorithmic maturity, and broad engineering applicability [5]. In classical implementations, the linear elastic state equations are predominantly discretized using standard conforming displacement-based Lagrange finite element methods (LFEM), where stress tensors are subsequently recovered within element interiors via numerical gradients of the discrete displacement fields. While such a displacement-driven paradigm is computationally efficient and sufficient for conventional compliance minimization problems, it encounters severe discretization errors and numerical bottlenecks in design problems governed by accurate local stress evaluation or nearly incompressible material behavior.

In local stress-constrained problems, constraints are imposed element-wise, whose quantity scales with the mesh size, and their values are strictly dictated by the accuracy of the discrete stress field. In standard displacement elements, differentiating displacement shape functions reduces the stress approximation accuracy by one order (e.g., linear elements yield only piecewise constant stresses), accompanied by non-physical discontinuous jumps in both normal and tangential stresses across element interfaces. If these post-processed discrete stresses are directly substituted into local constraints, the mutual coupling among discretization errors, smoothing strategies, and the optimization model readily induces local stress oscillations or traps the structural evolution in over-optimized designs [15, 16, 22, 23]. In nearly incompressible material regimes, as Poisson’s ratio approaches the incompressible limit ($\nu_0 \rightarrow 0.5$), the bulk modulus of plane-strain or three-dimensional continua tends to infinity relative to the shear modulus, imposing a stringent divergence-free deformation constraint ($\text{div} \mathbf{u} \approx 0$). Owing to the lack of sufficient deformation degrees of freedom in the discrete displacement space, standard low-order displacement elements suffer from severe volumetric locking [24], which artificially exaggerates structural stiffness and produces pathological thin members and spurious hinges during topological evolution [21, 25].

Mixed finite element methods offer an intrinsic variational pathway to overcome the aforementioned mechanical challenges [6]. Based on the Hellinger–Reissner two-field variational principle, the symmetric stress $\boldsymbol{\sigma}$ and displacement $\mathbf{u}$ are treated simultaneously as independent unknowns. The stress trial functions belong directly to the symmetric tensor $H(\text{div})$ space, naturally guaranteeing strict continuity of normal tractions across inter-element boundaries, while the divergence equilibrium equation is satisfied exactly in the weak form. Because the compliance bilinear form remains uniformly bounded in the incompressible limit, mixed elements satisfying the discrete inf-sup (Babuška–Brezzi) condition fundamentally circumvent volumetric locking [7, 26]. The truly-mixed topology optimization studies by Bruggi and Venini [9, 10] and Bruggi [8] demonstrated that directly discretizing stress and displacement provides an effective means for nearly incompressible structural design and stress-constrained optimization. However, most existing mixed finite element topology optimization literature is restricted to lowest-order discretizations, structured quadrilateral meshes, or specific hybridized formulations. For unstructured simplicial meshes (triangles and tetrahedra), constructing strictly symmetric and conforming $H(\text{div})$ spaces has long posed substantial mathematical difficulties, often necessitating additional multipliers to relax the symmetry into weakly symmetric formulations [1–3].

The Hu–Zhang mixed finite element family, introduced by Hu and Zhang [20] and Hu [18],

represents a landmark breakthrough in the construction of symmetric stress finite elements. By directly constructing full polynomial spaces $\mathbb{P}_k(\mathcal{S})$ on simplices, this method explicitly decouples normal trace and tangential components via a geometric entity decomposition, ensuring that inter-element normal tractions are uniquely governed by interface degrees of freedom, thereby establishing a family of strongly symmetric conforming spaces for arbitrary polynomial degrees. Building upon this foundation, a suite of essential theories has progressively emerged to facilitate its practical computation: Chen et al. [11] addressed the high polynomial degree requirement of the native formulation ($k \geq d+1$) by devising a symmetric matrix jump stabilization mechanism for lower orders ($k \leq d$); Hu and Ma [19] developed a local stress relaxation theory at complex boundary corners, eliminating over-constraint at geometric singularities; Chen et al. [12, 13] investigated fast auxiliary space preconditioners and residual-based a posteriori error estimators; and Chen et al. [14] subsequently formulated geometric decompositions and efficient algorithms for high-order simplicial face and edge elements. In light of the growing trend toward incorporating high-order finite elements into topology optimization [27], how to seamlessly embed such arbitrary-order symmetric stress approximations on simplicial meshes into density-based topology optimization—overcoming the design dependence introduced by non-homogeneous traction lifting, the corner stress relaxation amidst evolving structural boundaries, and the consistent adjoint sensitivity analysis driven by independent stresses—still lacks a systematic variational formulation and computational framework.

To address the aforementioned theoretical and algorithmic bottlenecks, this paper establishes an arbitrary-order Hu–Zhang mixed finite element framework for density-based topology optimization on simplicial meshes, built upon the open-source finite element library FEALPy [28] and the modular topology optimization platform SOPTX [17]. The principal contributions of this work are summarized as follows:

1. **Variational formulation and consistent adjoint analysis for simplicial mixed topology optimization:** Starting from the Hellinger–Reissner variational principle with non-homogeneous Neumann boundary conditions, an extrinsic traction lifting field is employed to handle boundary traction data, systematically revealing the design dependence of the load vector induced by the lifting field. Based on the reconstructed total stress, we derive a consistent adjoint sensitivity analysis framework for both complementary energy objectives and augmented Lagrangian local stress constraints.
2. **Consistent degree-of-freedom management and operator reuse for optimization iterations:** By casting the intricate high-order tensor degrees of freedom (DOFs) into a unified management of normal trace continuity, and integrating a global reference coordinate frame with local corner stress relaxation at geometric singularities, we construct an efficient computational implementation wherein the discrete divergence and stabilization operators are fully reused both structurally and numerically throughout the optimization process, requiring dynamic updates exclusively for the compliance blocks.
3. **Cross-order numerical evaluation and mechanical applicability boundaries:** We design a comprehensive benchmark suite spanning low- and high-order manufactured so-

lution convergence rates, cross-order compliance consistency between displacement and mixed formulations, volumetric locking alleviation in nearly incompressible regimes, and denominator-free local stress-constrained optimization. This objectively delineates the practical applicability boundaries of independent symmetric stress approximations relative to standard displacement elements in terms of topological evolution, stress fidelity, and computational cost.

The remainder of this paper is organized as follows. Section 2 introduces the governing equations of linear elasticity, the Hellinger–Reissner mixed variational formulation, and the traction lifting field at the continuum level. Section 3 elaborates on the tensor geometric decomposition on simplices, the construction of arbitrary-order conforming spaces, jump stabilization for lower polynomial degrees, local corner stress relaxation, and the assembly of discrete saddle-point systems. Section 4 establishes the two-parameter material interpolation, the complementary energy compliance minimization model, singularity-free apparent stress constraints, and consistent adjoint sensitivities based on the total stress. Section 5 presents convergence rate verifications with manufactured solutions alongside three representative benchmark problems in topology optimization. Finally, Section 6 summarizes the conclusions and delineates the operational boundaries of the proposed framework.

2 Preliminaries

2.1 Notation and Simplicial Mesh

Let $\Omega \subset \mathbb{R}^d$ ($d=2,3$) be a bounded Lipschitz polyhedral domain with boundary $\partial\Omega = \overline{\Gamma_D} \cup \overline{\Gamma_N}$, where $\Gamma_D \cap \Gamma_N = \emptyset$ and $|\Gamma_D| > 0$. Here Γ_D denotes the Dirichlet boundary where displacements are prescribed, and Γ_N represents the Neumann boundary subjected to surface tractions. Let $\mathbb{S}$ denote the space of real symmetric second-order tensors of order $d \times d$, and $\mathbb{I}$ be the second-order identity tensor.

For symmetric tensors $\sigma, \tau \in \mathbb{S}$, the Frobenius inner product and trace are denoted by $\sigma : \tau = \sum_{i,j=1}^d \sigma_{ij} \tau_{ij} = \text{tr}(\sigma \tau)$. For tensor field τ and vector field u , the divergence and symmetric gradient operators are defined by

$$(\text{div } \tau)_i = \sum_{j=1}^d \frac{\partial \tau_{ij}}{\partial x_j}, \quad \varepsilon(u) = \frac{1}{2} (\nabla u + \nabla u^\top). \quad (2.1)$$

Let $\mathcal{T}_h$ be a shape-regular conforming simplicial mesh of Ω (triangles for $d=2$ and tetrahedra for $d=3$), characterized by mesh size $h = \max_{T \in \mathcal{T}_h} h_T$. We denote the set of vertices by $\mathcal{V}_h$, edges by $\mathcal{E}_h$, and $(d-1)$ -dimensional faces by $\mathcal{F}_h$ (with $\mathcal{F}_h = \mathcal{E}_h$ in two dimensions). The face set is partitioned into internal faces $\mathcal{F}_h^i$, Dirichlet boundary faces $\mathcal{F}_h^D$, and Neumann boundary faces $\mathcal{F}_h^N$, such that $\mathcal{F}_h = \mathcal{F}_h^i \cup \mathcal{F}_h^D \cup \mathcal{F}_h^N$.

We adopt standard L^2 inner products and norms on Ω , denoted by $(\cdot, \cdot)_\Omega$ and $\|\cdot\|_{0,\Omega}$. The symmetric tensor $H(\text{div})$ space is defined as

$$H(\text{div}, \Omega; \mathbb{S}) = \left\{ \tau \in L^2(\Omega; \mathbb{S}) : \text{div } \tau \in L^2(\Omega; \mathbb{R}^d) \right\}, \quad (2.2)$$

equipped with the graph norm $\|\tau\|_{H(\text{div})}^2 = \|\tau\|_{0,\Omega}^2 + \|\text{div } \tau\|_{0,\Omega}^2$.

2.2 Linear Elasticity and Hellinger–Reissner Variational Formulation

Consider the small-strain isotropic linear elasticity boundary value problem:

$$\begin{aligned} -\text{div } \sigma &= \mathbf{b} & \text{in } \Omega, \\ \mathcal{A}\sigma &= \varepsilon(\mathbf{u}) & \text{in } \Omega, \\ \mathbf{u} &= \mathbf{u}_D & \text{on } \Gamma_D, \\ \sigma \mathbf{n} &= \mathbf{g} & \text{on } \Gamma_N, \end{aligned} \quad (2.3)$$

where $\sigma \in \mathbb{S}$ is the symmetric Cauchy stress tensor, $\mathbf{u}$ is the displacement vector, $\mathbf{b}$ is the body force, $\mathbf{g}$ is the prescribed surface traction, and $\mathbf{n}$ is the unit outward normal vector on $\partial\Omega$. The fourth-order compliance tensor $\mathcal{A}$ acts on any symmetric tensor $\tau \in \mathbb{S}$ as

$$\mathcal{A}\tau = \frac{1}{2\mu} \left(\tau - \frac{\lambda}{2\mu + d\lambda} \text{tr}(\tau) \mathbb{I} \right), \quad (2.4)$$

with Lamé parameters $\lambda = \frac{Ev}{(1+\nu)(1-2\nu)}$ and $\mu = \frac{E}{2(1+\nu)}$ for plane strain/3D, where E is Young's modulus and ν is Poisson's ratio.

To handle the traction boundary condition $\sigma \mathbf{n} = \mathbf{g}$ on Γ_N , we define the affine trial space and the homogeneous test space:

$$\Sigma_g = \{ \tau \in H(\text{div}, \Omega; \mathbb{S}) : \tau \mathbf{n} = \mathbf{g} \text{ on } \Gamma_N \}, \quad \Sigma_0 = \{ \tau \in H(\text{div}, \Omega; \mathbb{S}) : \tau \mathbf{n} = \mathbf{0} \text{ on } \Gamma_N \}. \quad (2.5)$$

The displacement space is chosen as $V = L^2(\Omega; \mathbb{R}^d)$. The two-field Hellinger–Reissner variational problem seeks $(\sigma, \mathbf{u}) \in \Sigma_g \times V$ such that

$$\begin{aligned} a(\sigma, \tau) + b(\tau, \mathbf{u}) &= \langle \mathbf{u}_D, \tau \mathbf{n} \rangle_{\Gamma_D}, \quad \forall \tau \in \Sigma_0, \\ b(\sigma, \mathbf{v}) &= -(\mathbf{b}, \mathbf{v})_\Omega, \quad \forall \mathbf{v} \in V, \end{aligned} \quad (2.6)$$

where the bilinear forms and boundary duality pairing are given by

$$a(\sigma, \tau) = \int_\Omega \mathcal{A}\sigma : \tau \, dx, \quad b(\tau, \mathbf{v}) = \int_\Omega (\text{div } \tau) \cdot \mathbf{v} \, dx, \quad \langle \mathbf{u}_D, \tau \mathbf{n} \rangle_{\Gamma_D} = \int_{\Gamma_D} \mathbf{u}_D \cdot (\tau \mathbf{n}) \, ds. \quad (2.7)$$

Because $\mathbf{u} \in L^2(\Omega; \mathbb{R}^d)$, no inter-element continuity is imposed on displacements. Meanwhile, as Poisson's ratio $\nu \rightarrow 0.5$, the compliance bilinear form $a(\cdot, \cdot)$ remains uniformly bounded, ensuring locking-free behavior in the continuous formulation [6].

2.3 Non-homogeneous Traction Lifting and Total Stress Formulation

In the mixed variational formulation (2.6), the traction condition $\sigma \mathbf{n} = \mathbf{g}$ acts as an essential boundary condition on the stress field. To enable discretization within a standard linear subspace, we

introduce a design-independent traction lifting field $\sigma_g \in \Sigma_g$ satisfying $\sigma_g \mathbf{n} = \mathbf{g}$ on Γ_N . The total stress is then decomposed into

$$\sigma = \sigma_0 + \sigma_g, \quad \sigma_0 \in \Sigma_0. \quad (2.8)$$

Substituting (2.8) into (2.6), the problem is recast as: find $(\sigma_0, \mathbf{u}) \in \Sigma_0 \times V$ such that

$$\begin{aligned} a(\sigma_0, \boldsymbol{\tau}) + b(\boldsymbol{\tau}, \mathbf{u}) &= \langle \mathbf{u}_D, \boldsymbol{\tau} \mathbf{n} \rangle_{\Gamma_D} - a(\sigma_g, \boldsymbol{\tau}), \quad \forall \boldsymbol{\tau} \in \Sigma_0, \\ b(\sigma_0, \mathbf{v}) &= -(\mathbf{b}, \mathbf{v})_\Omega - b(\sigma_g, \mathbf{v}), \quad \forall \mathbf{v} \in V. \end{aligned} \quad (2.9)$$

Once σ_0 is obtained, the total stress field $\sigma = \sigma_0 + \sigma_g$ is reconstructed and subsequently utilized for objective evaluation and stress constraint enforcement throughout the topology optimization procedure.

3 Arbitrary-Order Hu–Zhang Elements

3.1 Geometric Tangential–Normal Decomposition on Simplices

On a d -dimensional simplex $T \in \mathcal{T}_h$, let $\lambda_0, \dots, \lambda_d$ denote the standard barycentric coordinates associated with vertices $\mathbf{x}_0, \dots, \mathbf{x}_d$. For any face F_i opposite to vertex $\mathbf{x}_i$, the outward unit normal is $\mathbf{n}_i = -\nabla \lambda_i / \|\nabla \lambda_i\|$.

Following Hu and Zhang [20], the space of polynomial symmetric tensors $\mathbb{P}_k(T; \mathbb{S})$ can be characterized via a geometric entity decomposition into vertex, edge, face, and interior modes:

$$\mathbb{P}_k(T; \mathbb{S}) = \bigoplus_{\ell=0}^d \bigoplus_{e \in \Delta_\ell(T)} \mathcal{B}_{k,e}(T), \quad (3.1)$$

where $\Delta_\ell(T)$ denotes the set of ℓ -dimensional sub-simplices of T . On each sub-simplex e , polynomial tensor modes are split into normal and tangential components with respect to adjacent faces. Specifically, for any inter-element interface $F = T^+ \cap T^-$, normal trace continuity requires:

$$\boldsymbol{\tau}|_{T^+} \mathbf{n}_F = \boldsymbol{\tau}|_{T^-} \mathbf{n}_F \quad \text{on } F. \quad (3.2)$$

3.2 Arbitrary-Order Spaces and Standard Discrete Formulation

For an arbitrary integer polynomial order $k \geq 1$, the conforming Hu–Zhang finite element space for the symmetric stress tensor is defined globally as:

$$\Sigma_{k,h} = \{ \boldsymbol{\tau}_h \in H(\text{div}, \Omega; \mathbb{S}) : \boldsymbol{\tau}_h|_T \in \mathbb{P}_k(T; \mathbb{S}), \forall T \in \mathcal{T}_h \}. \quad (3.3)$$

The homogeneous subspace conforming to the traction-free boundary conditions is denoted by $\Sigma_{k,h}^0 = \Sigma_{k,h} \cap \Sigma_0$. The corresponding discrete displacement space is chosen as the piecewise polynomial space of degree $(k-1)$:

$$V_{k-1,h} = \left\{ \mathbf{v}_h \in L^2(\Omega; \mathbb{R}^d) : \mathbf{v}_h|_T \in \mathbb{P}_{k-1}(T; \mathbb{R}^d), \forall T \in \mathcal{T}_h \right\}. \quad (3.4)$$

For $k \geq d+1$, the pair $\Sigma_{k,h}^0 \times V_{k-1,h}$ satisfies the discrete inf-sup condition uniformly with respect to h , yielding optimal convergence:

$$\|\sigma - \sigma_h\|_{H(\text{div})} + \|\mathbf{u} - \mathbf{u}_h\|_0 \leq Ch^k (\|\sigma\|_{k+1} + \|\mathbf{u}\|_k). \quad (3.5)$$

3.3 Jump Stabilization for Low-Order Elements ($1 \leq k \leq d$)

For low-order discretizations ($k=1$ in 2D and $k=1,2$ in 3D), the standard inf-sup condition fails on general simplicial meshes. To restore stability while preserving optimal convergence, we employ the symmetric matrix jump stabilization proposed by Chen et al. [11].

For any internal face $F \in \mathcal{F}_h^i$ shared by elements T^+ and T^- , let $[\![\tau_h]\!] = \tau_h|_{T^+} - \tau_h|_{T^-}$ be the tensor jump across F . The stabilized bilinear form $a_s(\cdot, \cdot)$ is defined on $\Sigma_{k,h} \times \Sigma_{k,h}$ as:

$$a_s(\sigma_h, \tau_h) = a(\sigma_h, \tau_h) + \sum_{F \in \mathcal{F}_h^i} \gamma_0 h_F \int_F [\![\sigma_h]\!] : [\![\tau_h]\!] ds, \quad (3.6)$$

where $\gamma_0 > 0$ is a dimensionless stabilization parameter (taken as $\gamma_0 = 1.0$ in our implementations). Since the exact solution $\sigma \in H^1(\Omega; \mathbf{S})$ satisfies $[\![\sigma]\!] = \mathbf{0}$, this jump stabilization is strongly consistent.

3.4 Partial Relaxation of Vertex Continuity at Corners

At boundary vertices where boundary conditions abruptly switch (e.g., from Dirichlet to Neumann, or between non-collinear traction faces), strong C^0 tensor continuity at the vertex can cause severe artificial over-constraints (locking of stress tractions) [19].

Following Hu and Ma [19], we adopt a two-element local vertex relaxation technique: for any corner vertex $x_V \in \partial\Omega$ with discontinuity in normal directions or boundary condition types, the global tensor degrees of freedom associated with x_V are decoupled between adjacent boundary sub-domains. This partial relaxation eliminates non-physical stress singularities and restores the optimal convergence rates.

3.5 Discrete Saddle-Point System and Matrix Reuse

Let $\{\phi_i\}_{i=1}^{N_\sigma}$ and $\{\psi_j\}_{j=1}^{N_u}$ denote the basis functions of $\Sigma_{k,h}^0$ and $V_{k-1,h}$, respectively. The algebraic saddle-point system takes the form:

$$\begin{bmatrix} \mathbf{A}(\rho) & \mathbf{B}^\top \\ \mathbf{B} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{U}_\sigma \\ \mathbf{U}_u \end{bmatrix} = \begin{bmatrix} \mathbf{F}_\sigma(\rho) \\ \mathbf{F}_u \end{bmatrix}, \quad (3.7)$$

where $\mathbf{A}(\rho)_{ij} = a_s(\phi_j, \phi_i; \rho)$ depends on the design density vector ρ , while the divergence coupling matrix $\mathbf{B}_{ji} = b(\phi_i, \psi_j) = \int_\Omega (\text{div } \phi_i) \cdot \psi_j dx$ is purely geometric and strictly design-independent. Consequently, in iterative topology optimization, $\mathbf{B}$ is assembled only once during initialization and reused across all subsequent optimization cycles.

4 Density-Based Topology Optimization with Independent Stress

4.1 Design Variables, Filtering, Projection, and Material Interpolation

We parameterize the structural layout using an element-wise design variable vector $\boldsymbol{\rho} = \{\rho_e\}_{e=1}^{N_e} \in [0,1]^{N_e}$. To prevent mesh dependency and checkerboard instabilities, a standard linear density filter with filter radius $R_{\min}$ is applied, followed by a smoothed Heaviside threshold projection:

$$\tilde{\rho}_e = \frac{\sum_{i \in \mathcal{N}_e} w_{ei} v_i \rho_i}{\sum_{i \in \mathcal{N}_e} w_{ei} v_i}, \quad \bar{\rho}_e = \frac{\tanh(\beta\eta) + \tanh(\beta(\tilde{\rho}_e - \eta))}{\tanh(\beta\eta) + \tanh(\beta(1 - \eta))}, \quad (4.1)$$

where β is the projection sharpness parameter, $\eta \in (0,1)$ is the threshold, and v_i is the volume of element i .

Under the complementary energy setting, the compliance tensor $\mathcal{A}(\bar{\rho})$ is interpolated using a dual-parameter penalization:

$$\mathcal{A}(\bar{\rho}_e) = (\bar{\rho}_e^p + (1 - \bar{\rho}_e^p)\epsilon_\sigma)^{-1} \mathcal{A}_0, \quad (4.2)$$

where $p \geq 3$ is the penalization exponent, $\epsilon_\sigma = 10^{-6}$ is a small positive parameter preventing singularity in void regions, and $\mathcal{A}_0$ is the compliance tensor of the base solid material.

4.2 Complementary Energy Compliance Formulation

Using the reconstructed total stress $\boldsymbol{\sigma} = \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_g$, the structural compliance is naturally quantified via the complementary strain energy:

$$C(\bar{\rho}) = \frac{1}{2} \int_{\Omega} \mathcal{A}(\bar{\rho}) \boldsymbol{\sigma} : \boldsymbol{\sigma} dx = \frac{1}{2} a(\boldsymbol{\sigma}, \boldsymbol{\sigma}; \bar{\rho}). \quad (4.3)$$

The compliance minimization problem with a prescribed volume fraction $V_f \in (0,1)$ is formulated as:

$$\begin{aligned} \min_{\bar{\rho}} \quad & C(\bar{\rho}) = \frac{1}{2} a(\boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_g, \boldsymbol{\sigma}_0 + \boldsymbol{\sigma}_g; \bar{\rho}) \\ \text{s.t.} \quad & \mathbf{A}(\bar{\rho}) \mathbf{U}_\sigma + \mathbf{B}^\top \mathbf{U}_u = \mathbf{F}_\sigma(\bar{\rho}), \\ & \mathbf{B} \mathbf{U}_\sigma = \mathbf{F}_u, \\ & \frac{\sum_{e=1}^{N_e} v_e \bar{\rho}_e}{\sum_{e=1}^{N_e} v_e} - V_f \leq 0, \quad 0 \leq \rho_e \leq 1, \quad e = 1, \dots, N_e. \end{aligned} \quad (4.4)$$

4.3 Denominator-Free Relaxed Local Stress Constraints

For local stress-constrained problems, conventional formulations divide the microscopic stress by the penalized density ($\boldsymbol{\sigma} / \bar{\rho}^q$), creating severe numerical singularities when $\bar{\rho} \rightarrow 0$. To overcome this, we adopt a denominator-free relaxed apparent von Mises stress formulation.

The element apparent von Mises stress $\sigma_{\text{vM},e}$ is computed from the total stress σ :

$$\sigma_{\text{vM},e} = \left(\frac{1}{v_e} \int_{T_e} \frac{3}{2} \text{dev}(\sigma) : \text{dev}(\sigma) dx \right)^{1/2}, \quad (4.5)$$

where $\text{dev}(\sigma) = \sigma - \frac{1}{d} \text{tr}(\sigma) \mathbb{I}$. The local stress constraint is defined without division:

$$g_e(\sigma, \bar{\rho}) = \sigma_{\text{vM},e}(\sigma) - \bar{\rho}_e^q \sigma_{\text{max}} \leq 0, \quad e = 1, \dots, N_e, \quad (4.6)$$

where σ_{max} is the allowable material yield limit, and $q \in [0.5, 1]$ relaxes the allowable threshold in low-density void regions, completely eliminating the “stress singularity” phenomenon.

4.4 Augmented Lagrangian Formulation

To solve the large number of local stress constraints efficiently, we construct an Augmented Lagrangian (AL) functional:

$$\mathcal{L}(\rho, \lambda, \mu) = f_0(\bar{\rho}) + \sum_{e=1}^{N_e} \left[\lambda_e \max \left(g_e(\sigma, \bar{\rho}), -\frac{\lambda_e}{2\mu} \right) + \mu \max \left(g_e(\sigma, \bar{\rho}), -\frac{\lambda_e}{2\mu} \right)^2 \right], \quad (4.7)$$

where $f_0(\bar{\rho})$ is the objective function (e.g., structural volume fraction), λ_e is the Lagrange multiplier associated with element e , and $\mu > 0$ is the quadratic penalty parameter. Multipliers are updated in outer iterations via:

$$\lambda_e^{(k+1)} = \max \left(0, \lambda_e^{(k)} + 2\mu^{(k)} g_e(\sigma^{(k)}, \bar{\rho}^{(k)}) \right). \quad (4.8)$$

4.5 Consistent Adjoint Sensitivity Analysis

A crucial novelty of the present formulation is accounting for the design dependence embedded in the lifted right-hand side $\mathbf{F}_\sigma(\bar{\rho}) = -\mathbf{A}(\bar{\rho})\mathbf{U}_{\sigma_g}$.

For an arbitrary response functional $\Phi(\sigma_0, \bar{\rho})$, its total sensitivity with respect to projected density $\bar{\rho}_e$ is obtained by defining the Lagrangian:

$$\mathcal{L}_{\text{state}} = \Phi(\sigma_0, \bar{\rho}) - \lambda_\sigma^\top \left(\mathbf{A}(\bar{\rho})\mathbf{U}_\sigma + \mathbf{B}^\top \mathbf{U}_u - \mathbf{F}_\sigma(\bar{\rho}) \right) - \lambda_u^\top (\mathbf{B}\mathbf{U}_\sigma - \mathbf{F}_u). \quad (4.9)$$

The discrete adjoint system reads:

$$\begin{bmatrix} \mathbf{A}(\bar{\rho}) & \mathbf{B}^\top \\ \mathbf{B} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \lambda_\sigma \\ \lambda_u \end{bmatrix} = \begin{bmatrix} \frac{\partial \Phi}{\partial \mathbf{U}_\sigma} \\ \mathbf{0} \end{bmatrix}. \quad (4.10)$$

Special Case 1: Compliance without jump stabilization. For higher-order elements ($k \geq d+1$) without jump stabilization, the adjoint solution simplifies analytically to $\lambda_\sigma = \mathbf{U}_\sigma + \mathbf{U}_{\sigma_g}$, which recovers the explicit sensitivity:

$$\frac{\partial C}{\partial \bar{\rho}_e} = -\frac{1}{2} \int_{T_e} \frac{\partial \mathcal{A}}{\partial \bar{\rho}_e} \sigma : \sigma dx. \quad (4.11)$$

Special Case 2: Low-order stabilized formulation. For $1 \leq k \leq d$, the jump stabilization introduces inter-element couplings, requiring the full numerical solution of the discrete adjoint system (4.10). The sensitivities are then evaluated by chain rule with the density filter and Heaviside projection.

5 Numerical Results

In this section, we present comprehensive numerical experiments to demonstrate the accuracy, stability, and versatility of the arbitrary-order Hu–Zhang mixed finite element method for topology optimization. All simulations are implemented within the open-source library FEALPy and the topology optimization platform SOPTX.

5.1 Convergence Rate Verification via Manufactured Solutions

To verify the theoretical convergence rates of the arbitrary-order simplicial discretization and the jump stabilization, we consider a two-dimensional manufactured solution on the unit square domain $\Omega = (0,1)^2$. The exact displacement field is chosen as:

$$\mathbf{u}(x,y) = \begin{bmatrix} \sin(2\pi x) \sin(2\pi y) + \frac{x^3}{3} \\ \cos(2\pi x) \cos(2\pi y) + \frac{y^3}{3} \end{bmatrix}. \quad (5.1)$$

The exact stress tensor is obtained via the isotropic Hooke’s law $\boldsymbol{\sigma} = \mathcal{A}^{-1} \boldsymbol{\varepsilon}(\mathbf{u})$, with body force $\mathbf{b} = -\operatorname{div} \boldsymbol{\sigma}$. The domain is uniformly triangulated with mesh size h .

5.1.1 Higher-Order Convergence ($k=3,4$)

For higher-order discretizations without stabilization ($k=3$ and $k=4$), the mixed finite element pair $\boldsymbol{\Sigma}_{k,h} \times \mathbf{V}_{k-1,h}$ satisfies the inf-sup condition. The L^2 stress error $\|\boldsymbol{\sigma} - \boldsymbol{\sigma}_h\|_0$, $H(\operatorname{div})$ stress error $\|\boldsymbol{\sigma} - \boldsymbol{\sigma}_h\|_{H(\operatorname{div})}$, and L^2 displacement error $\|\mathbf{u} - \mathbf{u}_h\|_0$ exhibit optimal convergence orders $\mathcal{O}(h^{k+1})$, $\mathcal{O}(h^k)$, and $\mathcal{O}(h^k)$, respectively.

5.1.2 Lower-Order Stabilized Convergence ($k=1,2$)

For lower-order discretizations ($k=1$ and $k=2$), jump stabilization (3.6) is activated with $\gamma_0=1.0$. The numerical results demonstrate that the jump stabilization restores full stability and achieves optimal asymptotic convergence rates without introducing adverse numerical damping.

5.2 Benchmark Topology Optimization Cases

5.2.1 Compliance Minimization of Clamped Beam

We consider a standard 2D clamped beam of dimensions 2.0×1.0 discretized by unstructured triangular meshes. We compare the compliance minimization designs obtained using the Hu–Zhang mixed elements ($k=1,2,3$) against standard displacement-based Lagrange elements (P1, P2, P3).

In compressible regimes ($\nu = 0.3$), both formulations yield virtually identical topological layouts and objective values, confirming the mathematical consistency of the complementary energy formulation.

5.2.2 Nearly Incompressible Structural Optimization (2D Bearing)

To examine performance under the nearly incompressible limit, we investigate the classic 2D bearing benchmark with Poisson’s ratio approaching $\nu \rightarrow 0.4999$. Standard displacement-based low-order elements suffer severe volumetric locking, generating non-physical checkerboard artifacts and spurious slender struts. In contrast, the arbitrary-order Hu–Zhang elements completely circumvent locking, generating smooth, physically sound load-carrying architectures.

5.2.3 Local Stress-Constrained Cantilever Beam

We examine a 2D cantilever beam subjected to local von Mises stress constraints. Employing the denominator-free relaxed constraint formulation (4.6) and the augmented Lagrangian solver, the optimization converges smoothly without stress singularities. Compared with post-processed stresses in displacement models, the conforming normal traction of the Hu–Zhang elements yields strictly continuous boundary tractions and accurate stress evaluations, effectively eliminating stress concentration artifacts near re-entrant corners.

6 Concluding Remarks

6.1 Key Conclusions

In this paper, we have systematically established an arbitrary-order Hu–Zhang mixed finite element framework for density-based continuum topology optimization on simplicial meshes. By adopting the Hellinger–Reissner variational principle with independent symmetric Cauchy stress and displacement fields, the proposed methodology overcomes the primary bottlenecks of classical displacement-based models. The main conclusions are:

1. **Exact Normal Traction Continuity and Stress Fidelity:** The trial stress field naturally resides in $H(\text{div}, \Omega; \mathbb{S})$, guaranteeing inter-element normal traction continuity. This fundamentally avoids the stress degradation and inter-element jumps inherent in displacement post-processing, providing an accurate foundation for local stress constraints.
2. **Locking-Free Nearly Incompressible Design:** Because the compliance bilinear form remains uniformly bounded as $\nu \rightarrow 0.5$, the mixed formulation completely eliminates volumetric locking on general simplicial meshes, without requiring specialized selective integration or artificial pressure projections.
3. **Consistent Lifting and Adjoint Formulations:** By introducing design-independent traction lifting, non-homogeneous Neumann boundary conditions are seamlessly resolved, with the design-dependence of the resulting right-hand side rigorously accounted for in the adjoint sensitivity analysis.

6.2 Future Work and Outlook

Future research directions include:

1. Extending the arbitrary-order mixed topology optimization framework to full three-dimensional tetrahedral meshes;
2. Developing adaptive mesh refinement (AMR) and hp -adaptive strategies driven by Hu–Zhang mixed residual-based error estimators;
3. Integrating geometric nonlinearity and elastoplastic constitutive laws into the mixed topology optimization formulation.

A Explicit Basis and Corner Relaxation DOF Inventory

A.1 Local Tensor Basis and Vertex Degrees of Freedom for $k=1$

On a triangular element $T \in \mathcal{T}_h$, let $\{\lambda_0, \lambda_1, \lambda_2\}$ denote the barycentric coordinates. For polynomial degree $k=1$, the local polynomial space $\mathbb{P}_1(T; \mathbb{S})$ has dimension $\dim \mathbb{P}_1(T; \mathbb{S}) = 3 \times \frac{2 \times 3}{2} = 9$.

The local basis functions corresponding to each vertex x_i ($i=0,1,2$) are formed by tensor products of barycentric basis functions with symmetric rank-one matrices:

$$\phi_{i,\alpha\beta} = \lambda_i(\mathbf{t}_\alpha \otimes \mathbf{t}_\beta + \mathbf{t}_\beta \otimes \mathbf{t}_\alpha), \quad (\text{A.1})$$

where $\mathbf{t}_\alpha, \mathbf{t}_\beta$ are unit edge tangent vectors meeting at vertex x_i . Across an inter-element edge e , continuity of normal tractions $(\boldsymbol{\tau} \mathbf{n}_e)|_e$ is enforced through identifying corresponding normal-normal and normal-tangential trace components.

A.2 Corner Relaxation Degree of Freedom Mapping

At an exterior boundary corner vertex x_V , the normal vectors of the two incident boundary edges e_1, e_2 are linearly independent. If full continuity is enforced at x_V , the vertex stress tensor must satisfy both traction conditions simultaneously, leading to over-constraint when the prescribed boundary data or directions are non-collinear.

Under the two-element local corner relaxation strategy [19], the degrees of freedom associated with x_V are duplicated between the two adjacent boundary elements, decoupling the tangential-normal traces on e_1 and e_2 while preserving interior normal continuity.

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